

Technology Stack & Architecture Matrix

Date	23 September 2026
Team ID	
Project Name	Online Complaint Registration
Maximum Marks	4 Marks

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Interactive web interfaces for citizens, agents, and administrators.	React.js, JavaScript, HTML5, CSS3
2	Frontend Communication	Handles communication between frontend and backend APIs.	Axios
3	Backend Services	Handles API routes, business logic, and request processing.	Node.js, Express.js
4	Security Pipeline	Provides secure authentication and role-based access.	JWT, bcryptjs
5	Database	Stores users, complaints, roles, and assignment data.	MongoDB
6	Database Mapping	Provides schema-based data modeling and database operations.	Mongoose ODM
7	Runtime Environment	Provides the execution environment for the CivicCare application.	Node.js v22.23.1

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology Implementation
1	Open-Source Frameworks	Uses open-source technologies for frontend, backend, and database development.	MERN Stack (MongoDB, Express.js, React, Node.js)
2	Security Implementations	Provides secure authentication, password protection, and role-based access.	JWT Authentication, bcryptjs
3	Scalable Architecture	Separates frontend, backend, and database components for easier development and scaling.	React, Node.js, Express.js, MongoDB
4	Availability	Provides continuous access to complaint registration, management, and tracking features.	Node.js Server, MongoDB
5	Performance	Provides responsive interfaces and efficient communication between frontend and backend.	React, Axios, Express.js